Signed multichannel reconstruction on a factorial analytic cylinder domain

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Abstract

For the signed heat kernel of even order $p = 2m$, we construct a multichannel reconstruction formula on a specified Banach completion of dyadic polynomial cylinders. The coefficient fields are genuine Bochner L^1 limits of native predictable Taylor coefficients and have degree-independent bounds on the original observable norm. The centered observable pairing is the sum of $p-1$ bounded signed channels in the continuous test dual. The proof begins with an exact finite signed telescope, proves quantitative increment and adjacent-refinement estimates, and only then passes to the observable completion. A joint diagonal realizes every completed observable through actual signed polynomial sums. The pairing is indefinite; no positive path-space measure or stochastic isometry is asserted. Static Appell reconstruction, unique higher-order polynomial Stein weight, missing first polynomial coordinate dual and actual Schwartz-density duals are proved separately from the chronological formula. In fourth order, these dual densities propagate in L^2 at positive time and extend to derivative-of-Dirac functionals at zero in the pointwise Schwartz-test topology. We distinguish canonical coefficient uniqueness from uniqueness of channel synthesis. Exact variation/path-law, quartic-error, deterministic L^2-input and terminal-channel obstructions delimit stronger interpretations. A normalized polynomial sequence also proves nonclosability of the mean in compact-open path topology, including the Gaussian case; zero-start and all-start Schwartz testing are distinguished. All principal results are supported by a frozen Lean development. Novelty relative to earlier higher-order signed calculus remains unresolved.

1 Scope and normalization

Fix $m \in \mathbb{N}$, $m > 0$, a physical horizon $T > 0$, and put

$$p = 2m, \qquad c = (-1)^{m+1}.$$

The fundamental signed kernel k_τ, $\tau > 0$, has Fourier transform $e^{-\tau\xi^p}$ and generator $c\partial_x^p$. We normalize the Fourier transform by

$$k_\tau(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-\tau\xi^p} e^{-i\xi x}\, d\xi.$$

Changing the Fourier sign gives the same even real kernel. For each finite partition, independent increments mean the product of these signed densities; they do not mean independent random variables under an assumed positive path law.

The analytic domain $\mathcal{D}$ will be defined explicitly in Section 4. It has a dense injective polynomial core, a bounded signed mean $\mathcal{E}_T : \mathcal{D} \to \mathbb{R}$, and a bounded symmetric pairing

$$J_T : \mathcal{D} \longrightarrow \mathcal{D}^*, \qquad J_T(P)(G) = \mathcal{E}_T(PG)$$

on polynomial cylinders. Products in this last identity are finite polynomial products; bounded multiplication on the whole completion is not being assumed. Let λ denote Lebesgue measure on $[0,1]$, so time in the coefficient fields is normalized time and the physical horizon occurs in all kernel parameters and channel constants. Write $\mathcal{F} = L^1(\lambda; \mathcal{D}^*)$ for Bochner L^1.

Theorem 1.1. *There are canonical bounded linear maps $C_q : \mathcal{D} \to \mathcal{F}$, $q \geq 0$, satisfying*

$$\|C_q u\|_{\mathcal{F}} \leq \frac{\|J_T\|\, e^T}{q!}\, \|u\|_{\mathcal{D}}\,. \tag{1}$$

For $1 \leq q \leq p-1$, define

$$(I_q A)(v) = cpT(p-1)_{q-1} \int_0^1 A(s)(D_s^{p-q} v)\, ds, \tag{2}$$

where $(r)_k = r(r-1)\cdots(r-k+1)$ and D_s is the bounded cutoff derivative defined below. Then every $u \in \mathcal{D}$ satisfies

$$J_T(u - \mathcal{E}_T(u)1) = \sum_{q=1}^{p-1} I_q(C_q u) \qquad \text{in } \mathcal{D}^*. \tag{3}$$

On every fixed polynomial cylinder, C_q is the norm limit in $\mathcal{F}$ of its native predictable Taylor coefficient fields. Its values lie in the closed span of actual adapted polynomial step fields. Every completed observable admits a simultaneous polynomial/mesh approximation of all surviving fields and the right side of (3).

The equality is in the test dual, not an assertion of equality in $\mathcal{D}$ obtained by inverting J_T. Injectivity of J_T on the entire completion is not supplied. The theorem is therefore a signed Clark–Ocone-type reconstruction with its topology specified.

Higher-order signed heat calculus has substantial antecedents, including Hochberg's work [1]. Complex random-walk constructions give another generalized calculus [3], while time-changed pseudoprocesses are studied in [4]. The present paper does not claim the first higher-order Itô calculus. Its precise subject is the simultaneous native coefficient limit and bounded observable completion for these actual signed cylinder laws. Priority for this package remains unresolved.

2 Actual finite signed laws and polynomial conditioning

The function $e^{-\tau\xi^p}$ is Schwartz. Fourier inversion shows that k_τ is real, even and Schwartz, has integral one, and satisfies $k_\tau * k_\sigma = k_{\tau+\sigma}$. Thus polynomial functions are integrable against its total variation. Its moments are

$$\mu_r(\tau) = \int x^r k_\tau(x)\, dx = \begin{cases} (pk)!(c\tau)^k/k!, & r = pk, \\ 0, & p \nmid r. \end{cases} \tag{4}$$

For $r = 0$ this gives one. To obtain the formula, differentiate the Fourier transform at zero; Schwartz decay justifies differentiation under the integral. The derivative of $e^{-\tau\xi^p}$ vanishes unless the order is a multiple of p, and the resulting powers of i give c^k. Equivalently, for every polynomial f,

$$\int f(x) k_\tau(x)\, dx = (e^{c\tau\partial^p} f)(0). \tag{5}$$

The exponential series terminates on each polynomial.

An integration-by-parts identity that will control all power channels is

$$\int x f(x) k_\tau(x)\,dx = cp\tau \int f^{(p-1)}(x) k_\tau(x)\,dx. \tag{6}$$

It follows either from the moments (4), by testing monomials, or from differentiation of the Fourier transform. All boundary terms vanish by Schwartz decay.

On the dyadic grid with $N = 2^n$ cells set $h = 1/N$, $\tau = Th$, and use increment variables $x_0, \dots, x_{N-1}$. The signed expectation $\mathcal{E}_{T,n}$ integrates against $\prod_j k_\tau(x_j)\,dx_j$. The position variables are $y_i = x_0 + \cdots + x_i$. Refinement replaces each coarse increment by the sum of two fine increments. Convolution implies consistency of these actual signed expectations.

For $0 \le j \le N$, let $\Pi_j P$ integrate increments with indices $i \ge j$, leaving $x_0, \dots, x_{j-1}$ untouched. It is a polynomial in the retained increments. In particular $\Pi_0 P = \mathcal{E}_{T,n} P$, $\Pi_N P = P$, and $\Pi_j \Pi_k = \Pi_{\min(j,k)}$. These formulas follow from finite signed Fubini, with absolute integrability of polynomials against the product variation. The notation is polynomial conditioning; no general conditional expectation theorem for an indefinite path law is invoked.

3 Static Appell reconstruction and its actual density duals

Fix one kernel time $\tau > 0$ and abbreviate $a = c\tau \ne 0$. Let $L_a(P) = \int P(x) k_\tau(x)\,dx$ on real polynomials. Equation (4) says, as a *formal power series*,

$$L_a(e^{xz}) = e^{az^p}.$$

This notation means applying L_a separately to each polynomial coefficient. No exchange of an infinite exponential series with an integral is required. Define monic polynomials H_n by

$$e^{xz - az^p} = \sum_{n \ge 0} H_n(x) \frac{z^n}{n!}, \qquad H_n(x) = n! \sum_{j=0}^{\lfloor n/p \rfloor} \frac{(-a)^j x^{n-pj}}{j!(n-pj)!}. \tag{7}$$

Each sum is finite. Formal differentiation and the preceding moment identity give

$$H_0 = 1, \qquad H_n' = n H_{n-1}\ (n \ge 1), \qquad L_a(H_n) = \mathbf{1}_{n=0}.$$

Differentiating with respect to z also yields the raising identity $H_{n+1} = x H_n - p a H_n^{(p-1)}$, so these are exactly the source's pure-order Appell polynomials, not merely another monic family. The exponential-polynomial mechanism is classical. Hile and Stanoyevitch [6], Sections 3–4, construct broader heat polynomials by generating functions; their first-time-order case has exponential symbol $e^{tQ(z)}$, including the displayed family with $Q(z) = -az^p$ at time one. No new Appell construction is claimed.

Since H_n is monic of degree n, triangular elimination gives a unique finite expansion of every polynomial:

$$P = \sum_{n=0}^{\deg P} c_n(P) H_n, \qquad c_n(P) = \frac{L_a(P^{(n)})}{n!}. \tag{8}$$

Indeed $H_k^{(n)} = (k)_n H_{k-n}$ for $n \le k$, and is zero otherwise. Applying L_a leaves only $k = n$, with value $n!$, proving the coordinate formula and uniqueness without an orthogonality assumption.

Nondegenerate pairing does not supply polynomial coordinate duals

The signed polynomial pairing $[P,Q] = L_a(PQ)$ is nondegenerate. Here is an explicit witness for every nonzero Q of degree d:

$$[H_{(p-1)d}, Q] = \mathrm{lc}(Q)\,((p-1)d)!\,(pa)^d \neq 0. \tag{9}$$

To prove it, first use formal coefficients in

$$L_a(e^{xz-az^p}e^{xw}) = \exp\big(a((z+w)^p - z^p)\big)\,.$$

The coefficient of w^d, multiplied by $d!$, is a polynomial in z of degree at most $(p-1)d$. A product of factors with respective w-degrees $j_1,\dots,j_r \geq 1$, summing to d, has z-degree $pr - d \leq (p-1)d$. Equality forces $r = d$ and every $j_i = 1$. Thus its leading coefficient is $(pa)^d$. Comparing the coefficient of $z^n/n!$ proves

$$[H_n, x^d] = 0 \quad (n > (p-1)d), \qquad [H_{(p-1)d}, x^d] = ((p-1)d)!(pa)^d.$$

The lower monomials of Q vanish at the last index, proving (9), including $d = 0$.

This gives the unique polynomial higher-order Stein weight:

$$L_a\left(P\frac{x}{pa}\right) = L_a(P^{(p-1)}).$$

Existence is (6). If another polynomial W has this identity for all P, then $[P, W - x/(pa)] = 0$ for all P, and (9) forces $W = x/(pa)$.

In contrast, for $p > 2$ there is no polynomial Q with $[H_n, Q] = \mathbf{1}_{n=1}$ for all n. If such Q were zero the $n = 1$ test would fail. Otherwise put $d = \deg Q$ and test at $n = (p-1)d$. This index cannot equal one, since $p - 1 \geq 2$; the requested pairing is zero, contradicting (9). In particular there is no polynomial first-order weight W with

$$L_a(PW) = L_a(P') \qquad \text{for every polynomial } P. \tag{10}$$

Such a weight would represent c_1, since $L_a(H_n') = \mathbf{1}_{n=1}$. The Gaussian case $p = 2$ is excluded from this obstruction: its familiar weight $x/(2a)$ does satisfy the first-order identity. Nondegeneracy tests whether a polynomial is annihilated by *all* pairing probes; it does not assert that every linear coordinate functional is represented by a polynomial weight.

Actual Schwartz-density coordinate duals

There is nevertheless a concrete dual for every coordinate. Set

$$\rho_n(x) = \frac{(-1)^n}{n!}k_\tau^{(n)}(x). \tag{11}$$

These are real Schwartz functions, so $P\rho_n$ is absolutely Lebesgue integrable for every polynomial P. Repeated ordinary integration by parts, with vanishing boundary terms, gives

$$\int P(x)\rho_n(x)\,dx = c_n(P), \qquad \int H_j(x)\rho_n(x)\,dx = \mathbf{1}_{j=n}. \tag{12}$$

The $(-1)^n$ from integration by parts cancels that in (11). Thus the polynomial basis has actual Schwartz-density biorthogonal coordinates, even though its first coordinate has no polynomial dual

in the original signed pairing when $p > 2$. In particular ρ_1 cannot equal Wk_τ almost everywhere for any polynomial W: such equality would imply (10) after integrating every polynomial test. This statement excludes polynomial weights only. It supplies no $L^2(|k_\tau|\,dx)$ score bound, bounded completion of all coordinate maps, positive covariance or probability interpretation.

Finally define the centered polynomial primitive by

$$\mathcal{I}P = \sum_n \frac{c_n(P)}{n+1} H_{n+1}.$$

It satisfies $(\mathcal{I}P)' = P$ and $L_a(\mathcal{I}P) = 0$. Any other such primitive differs by a constant with mean zero, hence by zero. Testing on the basis gives, for every integer $r \geq 0$,

$$\mathcal{I}^r P^{(r)} = P - \sum_{n<r} c_n(P) H_n.$$

For $n \geq r$ the falling derivative factors cancel the successive primitive denominators; for $n < r$ the derivative is zero. The omitted part is all low Appell coordinates, rather than just the constant. For example $c_1(x^{p+1}) = (p+1)p!a \neq 0$, so reconstruction from the pth derivative by this primitive does not equal mere centering. These are finite polynomial identities. The primitive is not declared to be a chronological stochastic integral; the multitime analytic completion and native coefficient limits still require the separate proofs below.

Source correspondence: `PureOrderAppellCalculus`, `PureOrderAppellExpansion`, `PureOrderAppellPairing`, `PureOrderSteinUniqueness`, `PureOrderAppellPrimitive`, `PureOrderAppellTail`, `HigherHeatAppellDuals`.

Fourth-order propagation and the singular initial dual

Here specialize to $p = 4$, so the generator is $-\partial_x^4$ and the Appell parameter at time $t > 0$ is $a = -t$. Write k_t for this kernel and $\rho_{n,t} = (-1)^n \partial_x^n k_t / n!$. Let S_t be the Fourier multiplier $e^{-t\xi^4}$ on Lebesgue $L^2(\mathbb{R})$, or on Schwartz functions when appropriate. The Fourier normalization is the angular one fixed in Section 1; the formal sources instead use 2π frequency, with the same physical operator. All distribution pairings below are complex linear, without conjugating the test function.

Proposition 3.1. *For $s \geq 0$, $T > 0$ and $n \geq 0$,*

$$S_s \rho_{n,T} = \rho_{n,T+s}, \qquad \|\rho_{n,T+s}\|_2 \leq \|\rho_{n,T}\|_2 .$$

The map $s \mapsto \rho_{n,T+s}$ is continuous in L^2 and its right derivative at $s = 0$ is $-\partial_x^4 \rho_{n,T}$. There is also a continuous extension to $t = 0$ as tempered distributions with the topology of pointwise evaluation on Schwartz tests:

$$R_{n,t}(\varphi) = \frac{1}{n!} \int_{\mathbb{R}} k_t(x) \varphi^{(n)}(x)\, dx \quad (t > 0), \qquad R_{n,0}(\varphi) = \frac{\varphi^{(n)}(0)}{n!}. \tag{13}$$

For positive time, $R_{n,t}$ is the distribution of the actual Schwartz density $\rho_{n,t}$. For every fixed Schwartz test, including the right derivative at zero,

$$\frac{d}{dt} R_{n,t}(\varphi) = -\frac{1}{n!} \int k_t(x) \varphi^{(n+4)}(x)\, dx \quad (t > 0), \qquad R'_{n,0+}(\varphi) = -\frac{\varphi^{(n+4)}(0)}{n!}. \tag{14}$$

Proof. Fourier transformation sends ∂_x^n to multiplication by $(i\xi)^n$. It therefore commutes with S_s, and multiplying $e^{-s\xi^4}$ and $e^{-T\xi^4}$ gives the first identity. Plancherel and $|e^{-s\xi^4}| \le 1$ give the contraction. For each fixed T, the Fourier transform of $\rho_{n,T}$ is a polynomial times $e^{-T\xi^4}$. Dominated convergence proves strong continuity, using $4|\widehat{\rho}_{n,T}|^2$ as a squared majorant. For the right derivative use

$$\left|\frac{e^{-s\xi^4}-1}{s}\right| \le \xi^4 \quad (s>0).$$

The squared majorant $\xi^8|\widehat{\rho}_{n,T}|^2$ is integrable, and the multiplier limit is $-\xi^4$. This identifies the derivative.

Repeated integration by parts gives $\int \varphi\rho_{n,t} = \int k_t\varphi^{(n)}/n!$, with no boundary terms because both functions are Schwartz. For any Schwartz g, Fourier inversion and evenness of the multiplier give

$$\int k_t g = \frac{1}{2\pi}\int e^{-t\xi^4}\widehat{g}(\xi)\,d\xi \longrightarrow g(0).$$

The integrable majorant is $|\widehat{g}|$, independently of t. This proves continuity of every evaluation in (13). Each evaluation at zero is a continuous Schwartz functional, as follows from the defining derivative seminorms. Thus the extension has exactly the asserted pointwise distribution topology. No stronger dual topology is required here.

Finally the difference quotients of the last Fourier integral are dominated by $\xi^4|\widehat{g}(\xi)|$ at zero and at any positive time. Differentiating gives the mean of $-g^{(4)}$. Take $g=\varphi^{(n)}$ and divide by $n!$ to obtain (14), with the displayed signs. □

The initial functional is $(-1)^n\delta_0^{(n)}/n!$, since a distributional derivative acts by $\delta_0^{(n)}(\varphi) = (-1)^n\varphi^{(n)}(0)$. It is not an initial L^2 density. For clarity, the following quantitative consequence is a human Fourier calculation, rather than an additional mapped Lean theorem. Plancherel in angular frequency gives

$$\|\rho_{n,t}\|_2^2 = \frac{1}{2\pi(n!)^2}\int_{\mathbb{R}} \xi^{2n}e^{-2t\xi^4}\,d\xi = C_n t^{-(2n+1)/4}, \qquad C_n = \frac{1}{2\pi(n!)^2}\int_{\mathbb{R}} y^{2n}e^{-2y^4}\,dy > 0. \tag{15}$$

The change of variables is $y = t^{1/4}\xi$; the last integral is finite by exponential decay and positive on an interval avoiding zero. Hence these densities have unbounded L^2 norm as $t \downarrow 0$, although every Schwartz evaluation converges. One can also see directly why the endpoint cannot be represented by an L^2 function: choose a smooth compactly supported ψ with $\psi^{(n)}(0) \neq 0$ and put $\psi_\varepsilon(x) = \varepsilon^n\psi(x/\varepsilon)$. Its nth derivative at zero is fixed, whereas $\|\psi_\varepsilon\|_2 = \varepsilon^{n+1/2}\|\psi\|_2 \to 0$, contradicting the Cauchy–Schwarz bound for such a representation.

This is deterministic evolution of coordinate densities and test functionals. It does not give bounded coordinate maps on the completed cylinder domain, a positive path law, or a chronological stochastic integral. The underlying fourth-order Fourier fundamental solution is classical; for instance, D'Ovidio [5], equations (4.1)–(4.2), gives this equation and its inverse Fourier kernel. We use neither that paper's subsequent multiplication assertions nor its fractional time-change results. No novelty is claimed for derivative propagation or the scaling calculation.

Source correspondence: `QuarticDualSemigroup.density_evolve`, `density_l2_contraction`, `density_l2_continuous`, `density_right_generator`. `QuarticDualDistributions.realized_apply`, `realized_continuous`, `realized_observable_generator`.

4 The common factorial domain

Let S be the union of the positive dyadic grid times in $(0, 1]$. A cylinder is a polynomial in finitely many formal position coordinates Y_s, $s \in S$. The direct-limit construction using increment refinement gives exactly this polynomial algebra: old positions are retained at their original times, and the cumulative coordinate change is invertible on each finite grid.

For a finite multiindex $\alpha : S \to \mathbb{N}$, let $|\alpha| = \sum_s \alpha_s$. For a polynomial $P = \sum_\alpha a_\alpha Y^\alpha$, define

$$\|P\|_{\mathcal{D}} = \sum_\alpha |\alpha|! \, |a_\alpha|. \tag{16}$$

This is the factorial of total degree, not the product of coordinate factorials. The coefficients here are in *position* coordinates, after converting a given increment polynomial by the inverse cumulative substitution. Refinement is an isometry because it only retains the same position labels. It also does not increase total degree. The norm is nondegenerate because polynomial coefficients are unique.

Define $\mathcal{D}$ as the completion of this normed polynomial space. More concretely, it is the closed span of the vectors $(|\alpha|!a_\alpha)_\alpha$ in ℓ^1 indexed by finitely supported time multiindices. The Lean construction uses all times in $(0, 1]$ as an ambient index and takes the closed span of the dyadic core. This yields the same space just described. It is a specified analytic coefficient completion, not a space of all bounded smooth path observables.

Lemma 4.1. *For any direction $v : S \to \mathbb{R}$ with $|v_s| \le 1$, the polynomial directional derivative $\partial_v = \sum_s v_s \partial_{Y_s}$ extends to $\mathcal{D}$ with norm at most one. Position substitutions that send each variable to another variable or zero, without scaling by a coefficient of modulus greater than one, are contractions. The finite signed projection Π_j has norm at most e^T on each cylinder norm.*

Proof. For a monomial of degree $d > 0$, the ratio of the weighted norm of its directional derivative to $d!$ is at most $\sum_s \alpha_s/d = 1$. The degree-zero derivative vanishes. Summation proves the derivative bound. Substitutions preserve total degree or lower it; merging equal monomials can only reduce the sum of absolute coefficients.

In position coordinates, differentiation in the increment x_j is the direction which is one on positions after that increment and zero before it. Its norm is therefore at most one. Integrating one increment is heat evolution $e^{c\tau\partial_{x_j}^p}$ followed by setting that increment to zero. Setting it to zero merges later positions with their predecessors, so is contractive. The heat operator has norm at most e^τ by its exponential series. Composing future heat operators has norm at most $e^{\sum \tau} \le e^T$, and the remaining clamping substitution is contractive. □

Taking the empty-past projection gives a bounded signed mean $\mathcal{E}_T$ on $\mathcal{D}$, with $\|\mathcal{E}_T\| \le e^T$ and $\mathcal{E}_T 1 = 1$. The cutoff derivative in (2) is

$$D_s = \partial_{v(s)}, \qquad v(s)_t = 1_{\{s \le t\}}. \tag{17}$$

It is contractive, as are all its powers. Applied to a fixed polynomial, it is a finite step function of s. Approximation by core polynomials shows $s \mapsto D_s^r u$ is strongly measurable for every $u \in \mathcal{D}$: the applied fields converge pointwise, even uniformly in the norm bound. This gives a bounded map into $L^\infty(\lambda; \mathcal{D})$.

A genuine initial-time gap in the coefficient norm

Bounded causal projections do not supply strong continuity at time zero in this particular completion. Write $s_n = 2^{-n}$ and let P_n be the completed signed projection onto the first cell of the level-n grid; its physical cutoff is Ts_n. Let $P_0^{\mathrm{empty}} u = (\mathcal{E}_T u)1$, where the superscript distinguishes the empty past from the level-zero first-cell projection. All these maps are bounded by e^T. For the single fixed terminal cylinder $U = Y_1$, we have

$$P_n U = Y_{s_n}, \qquad P_0^{\mathrm{empty}} U = 0, \qquad \left\| (P_n - P_0^{\mathrm{empty}}) U \right\|_{\mathcal{D}} = 1, \qquad \|U\|_{\mathcal{D}} = 1. \tag{18}$$

Here and below polynomial cylinders are identified with their canonical embeddings into $\mathcal{D}$.

Every identity has an elementary finite-law proof. On the level-n grid the terminal position is the sum of all increments. The projection retains the first increment and integrates the others. Each future increment has signed first moment zero by (4), and all its polynomial pairings are integrable, so the result is the first position Y_{s_n}. The empty projection integrates the terminal linear position to zero. The coefficient norm (16) of any one position variable is exactly $1! \cdot 1 = 1$, independent of its time label. Refinement retains that label and is isometric. Thus the identities persist in the completed operators, whose values on core cylinders agree with the original projections.

It follows from (18) that $P_n U$ does not converge to the empty-past value in $\mathcal{D}$. In particular P_n fails to converge strongly to P_0^{empty}, and the operator norm obeys

$$\left\| P_n - P_0^{\mathrm{empty}} \right\| \geq 1.$$

Nevertheless, for every fixed continuous zero-start path ω, $Y_{s_n}(\omega) = \omega(s_n) \to 0$. This is pointwise path evaluation, not convergence in the factorial coefficient norm. The same gap holds at order $p = 2$; it is not a failure caused only by signed laws. Under the separate Brownian realization with generator ∂_x^2, the first-position law has second moment $2Ts_n$, so its actual Brownian L^2 norm tends to zero as well. That last norm comparison is a human consequence of the variance convention and does not identify the whole coefficient completion with Brownian L^2.

The coefficient-norm proof is in `HigherHeatAnalyticTimeObstruction`.

The projection gap is in `HigherHeatCausalTimeObstruction`.

The terminal formula is in `HigherHeatTerminalPositions`.

The empty-past extension is in `HigherHeatCompletedEmptyPast`. This is a precise topology obstruction for these operators, not an obstruction to the native multichannel reconstruction proved below, whose limits have their separately stated dual and Bochner topologies.

An actual arithmetic transform outside the one-coordinate realization

The domain restriction can be tested on another verified family in the same library. In one position coordinate, the norm (16) is $\sum_{n\geq 0} n!|c_n|$. Set $u_n = n!c_n$. The corresponding one-coordinate function realization is

$$f_u(x) = \sum_{n\geq 0} u_n \frac{x^n}{n!}, \qquad u \in \ell^1(\mathbb{N}; \mathbb{R}). \tag{19}$$

This is the original coefficient realization, not a replacement Sobolev or weighted L^2 domain. On $|x| \leq R$, the series and each differentiated series converge uniformly: for the qth derivative, the sum of absolute terms is at most $e^R \sum_{n\geq q} |u_n|$. Termwise differentiation is therefore valid, and

$$f_u^{(q)}(0) = u_q, \qquad \sum_{q\geq 0} |f_u^{(q)}(0)| < \infty.$$

This condition is much stronger than being entire.

For a concrete exclusion, let $m \geq 1$ be an integer, $t \in \mathbb{R}$, and use the actual arithmetic theta kernel

$$K(v) = \sum_{j \geq 1} (2\pi^2 j^4 e^{9v} - 3\pi j^2 e^{5v}) e^{-\pi j^2 e^{4v}} \quad (v \geq 0), \qquad F_{m,t}(x) = \int_0^\infty e^{tv^{2m}} K(v) \cos(xv)\, dv.$$

The kernel is continuous and strictly positive: each summand factors as

$$\pi j^2 e^{5v} (2\pi j^2 e^{4v} - 3) e^{-\pi j^2 e^{4v}} > 0.$$

The series converges uniformly on compact intervals. At infinity it is bounded by $C \exp(Cv - ce^{4v})$, so every moment remains integrable after multiplication by $e^{tv^{2m} + Rv}$ for every fixed m, t, R. These bounds follow by separating half of the decaying exponential to sum the polynomial j factors. They justify all differentiations of the integral, including

$$F_{m,t}^{(2n)}(0) = (-1)^n \int_0^\infty v^{2n} e^{tv^{2m}} K(v)\, dv. \tag{20}$$

Continuity and positivity give $d = \min_{2 \leq v \leq 3} e^{tv^{2m}} K(v) > 0$. Hence the interval of length one supplies the lower bound

$$|F_{m,t}^{(2n)}(0)| \geq \int_2^3 v^{2n} d\, dv \geq d\, 4^n.$$

These derivative magnitudes do not even tend to zero. Thus no $u \in \ell^1$ can satisfy $f_u = F_{m,t}$ on the real line, since equality of the smooth functions would identify all their derivatives at zero, contradicting (19). The exact frozen theorem also records the bridge from complex flow derivatives to this real-function derivative conclusion.

Consequently the bounded derivative and heat operators on this one-coordinate factorial realization cannot simply be applied to these integer-order arithmetic transforms. This is a concrete domain obstruction, not a counterexample to the multichannel reconstruction theorem. It neither asserts injectivity of every completed product pairing nor excludes a different representation in another space or quotient. No arithmetic zero conclusion or threshold relation follows from this comparison. The coefficient growth mechanism is elementary; no novelty is claimed for it.

4.1 A concrete comparison with Fourier-measure tests

Let $\mathcal{A}$ denote the faithful real function range in (15), with its coefficient norm $\sum_n |u_n|$. Let $\mathcal{F}$ be the real-valued functions representable as

$$f(x) = \int_{\mathbb{R}} e^{i\xi x}\, d\mu(\xi), \qquad \int_{\mathbb{R}} (1 + |\xi|)^k\, d|\mu|(\xi) < \infty \quad \text{for every integer } k \geq 0,$$

where μ is a complex measure. This is the one-coordinate Fourier-measure test class used in Nishioka's construction [2, Section 2], restricted here to real-valued outputs. These statements compare actual functions on the whole real line.

Proposition 4.2. *Neither $\mathcal{A}$ nor $\mathcal{F}$ contains the other. More precisely, $x \mapsto x$ belongs to $\mathcal{A} \setminus \mathcal{F}$, and $x \mapsto \cos(\xi x)$ belongs to $\mathcal{F}$ for every real ξ, but belongs to $\mathcal{A}$ exactly when $|\xi| < 1$. In the latter case its coefficient norm is $(1 - \xi^2)^{-1}$.*

Proof. The sequence $u_1 = 1$, $u_n = 0$ otherwise realizes the polynomial x and has norm one. Every function in $\mathcal{F}$ is bounded by $|\mu|(\mathbb{R})$, directly from the integral and $|e^{i\xi x}| = 1$. Thus the unbounded polynomial is not in $\mathcal{F}$. For any fixed ξ, the finite atomic measure $(\delta_\xi + \delta_{-\xi})/2$ has every required weighted moment and represents $\cos(\xi x)$. Its derivatives at zero are zero in odd orders and $(-1)^j \xi^{2j}$ in order $2j$. Faithful derivative recovery therefore forces these exact coefficients for any realization in $\mathcal{A}$. Their absolute sum is $\sum_{j\geq 0} |\xi|^{2j}$, finite exactly when $|\xi| < 1$, with the claimed sum. When finite, the resulting factorial series is the ordinary cosine series, proving sufficiency as well as necessity. □

The frequency threshold belongs to this fixed coefficient norm; it is not invariant under a rescaling of that norm. This elementary comparison rules out either identity inclusion for these faithful one-coordinate function spaces. It does not rule out other multivariate presentations, quotient representations or bounded maps on smaller common domains. In particular it neither identifies the two weak reconstruction topologies nor proves priority for the completed multichannel theorem.

5 The product pairing and its signed character

Ordinary multiplication does not have norm at most the product of the norms in (16). Instead define on polynomials

$$B(P, Q)(Y) = P(Y/2)Q(Y/2).$$

For monomials of degrees d, e, its weighted norm ratio is $2^{-d-e}(d + e)!/(d!e!) \leq 1$. Summing absolute coefficients gives $\|B(P, Q)\| \leq \|P\| \, \|Q\|$, so B extends as a bounded symmetric bilinear map $\mathcal{D} \times \mathcal{D} \to \mathcal{D}$. Set

$$J_T(u)(v) = \mathcal{E}_{2^p T}\big(B(u, v)\big). \tag{21}$$

Then $\|J_T\| \leq e^{2^p T}$. Homogeneity of the moments shows that half-dilation at horizon $2^p T$ restores horizon T: $\mu_r(2^p\tau)2^{-r} = \mu_r(\tau)$. Applying this to each increment monomial proves

$$J_T(P)(G) = \mathcal{E}_T(PG) \tag{22}$$

for every pair of cylinders, and $J_T(u)(1) = \mathcal{E}_T u$ by continuity. This construction supplies the bounded product *pairing* without assuming bounded ordinary multiplication.

There is an explicit obstruction to any bounded ordinary-product extension. Fix one dyadic coordinate $Y = Y_s$ and let $P_n = Y^n/n!$. By (16),

$$\|P_n\|_{\mathcal{D}} = 1, \qquad \|Y\|_{\mathcal{D}} = 1, \qquad \|YP_n\|_{\mathcal{D}} = n + 1.$$

Thus even multiplication by this fixed polynomial Y has no bounded linear extension agreeing with the polynomial products. In particular no bounded bilinear map on $\mathcal{D} \times \mathcal{D}$ can agree with ordinary multiplication on every polynomial pair.

The bounded replacement B has a different limitation: it is commutative but not associative. On the same faithful polynomial core,

$$B(1, 1) = 1, \qquad B(1, Y) = Y/2,$$

and consequently

$$B(B(1, 1), Y) = Y/2 \neq Y/4 = B(1, B(1, Y)).$$

The two outputs are distinct since their coefficient norm difference is $1/4$. These are direct human consequences of the printed core formulas, not new named formal theorems. They rule out treating this damped operation as a Banach-algebra product. The construction of J_T uses B once and retains the explicit horizon correction; it requires no associativity of B.

`HigherHeatCommonDampedProduct.product_core`.

`HigherHeatCommonDampedProduct.product_bound`.

`HigherHeatCommonDampedProduct.product_comm`.

For $p = 4$ one has $c = -1$, so (4) gives

$$J_T(Y_1^2)(Y_1^2) = \mathcal{E}_T(Y_1^4) = -24T < 0.$$

Thus the pairing is indefinite. In particular the notation $\mathcal{E}_T$ cannot be used to justify positive Cauchy–Schwarz, martingale isometries, or Hilbert orthogonal projections.

For later estimates it is useful to name

$$W_j(H)(s) = 1_{(jh,(j+1)h]}(s) J_T(H).$$

This is an actual element of $\mathcal{F}$, with $\|W_j(H)\|_{\mathcal{F}} = h \|J_T(H)\|$. An adapted step field has this form summed over j, with H_j a polynomial depending only on increments before cell j. Let $\mathcal{A}_T$ be the closed linear span of all such fields in $\mathcal{F}$. This definition is the exact adaptedness space used below; it does not presuppose a probability filtration.

6 The bounded channel operators

For $A \in \mathcal{F}$ and $v \in \mathcal{D}$, the integrand $A(s)(D_s^r v)$ is integrable because its absolute value is at most $\|A(s)\| \, \|v\|$. Strong measurability follows from the bounded bilinear evaluation map and the applied-field measurability proved above. Define

$$Q_r(A)(v) = \int_0^1 A(s)(D_s^r v)\, ds.$$

Then $Q_r : \mathcal{F} \to \mathcal{D}^*$ is bounded with norm at most one. In addition to the surviving channels in (2), use the same formula for $q = p$. In that case

$$I_p = cp!TQ_0. \tag{23}$$

It is a drift channel, which will cancel exactly after centering. For $1 \le q \le p$,

$$\|I_q A\| \le pT(p-1)_{q-1} \, \|A\|_{\mathcal{F}}. \tag{24}$$

For recursive calculations put $K_k = cpT\binom{p-1}{k} Q_{p-1-k}$, $0 \le k < p$. Thus $I_q = (q-1)!K_{q-1}$. The factorial distinction is needed: the native Taylor coefficient contains $1/q!$, whereas conversion of a power channel contributes $(q-1)!$. Losing either factor changes the reconstruction formula.

7 Exact finite predictable innovations

Fix a grid and a polynomial P of total degree at most d. For cell j, put $Q_j = \Pi_{j+1} P$, and let R_j set $x_i = 0$ for $i \ge j$. Define the native coefficients

$$a_{j,q}(P) = \frac{1}{q!} R_j \partial_{x_j}^q Q_j. \tag{25}$$

They depend only on strict-past increments, vanish for $q > d$, and Lemma 4.1 gives

$$\|a_{j,q}(P)\|_{\mathcal{D}} \le \frac{e^T}{q!}\,\|P\|_{\mathcal{D}}\,. \tag{26}$$

Only the current and past increments survive in Q_j. Its exact Taylor expansion in the current increment therefore is $Q_j = \sum_{q=0}^{d} a_{j,q}(P)x_j^q$. Integrate that increment to obtain $\Pi_j P = \sum_{q=0}^{d} a_{j,q}(P)\mu_q(Th)$. Subtracting proves the cell identity

$$\Pi_{j+1}P - \Pi_j P = \sum_{q=1}^{d} a_{j,q}(P)\big(x_j^q - \mu_q(Th)\big). \tag{27}$$

The zeroth term vanishes because $\mu_0 = 1$. Summing over cells telescopes exactly:

$$P - \mathcal{E}_T P = \sum_{q=1}^{d}\sum_{j=0}^{N-1} a_{j,q}(P)\big(x_j^q - \mu_q(Th)\big). \tag{28}$$

No term has been discarded on a scaling heuristic. Every integral in this derivation is an actual finite signed polynomial integral.

Refine a fixed base polynomial P to successive grids and denote its refinement by P_n. Its position-coordinate norm is unchanged and its total degree stays at most d. Define the native field

$$A_{n,q}(P) = \sum_j W_j(a_{j,q}(P_n)). \tag{29}$$

Since cell lengths sum to one, (26) gives

$$\|A_{n,q}(P)\|_{\mathcal{F}} \le \frac{\|J_T\|\,e^T}{q!}\,\|P\|_{\mathcal{D}}\,. \tag{30}$$

This estimate is independent of degree and mesh, but does not by itself prove that the sequence converges. That requires the next two quantitative arguments.

8 Signed power estimates in the test dual

For strict-past H and integer $r \ge 0$, signed integration by parts gives the exact dual identity

$$J_T(Hx_j^{r+1}) = \sum_{k=0}^{p-1}(r)_k K_k\big(W_j(Hx_j^{r-k})\big). \tag{31}$$

Terms with $k > r$ vanish and can be omitted. To check the identity, test first against a polynomial G on a common finer grid. The coarse increment is the sum of fine increments in its cell. Apply (6) to each fine increment and use Leibniz's rule on $x_j^r G$. Derivatives of H vanish because it is strict past. The k derivatives on x_j^r give $(r)_k x_j^{r-k}$, the other derivatives give $D_s^{p-1-k}G$, and summing the fine durations gives exactly the cell integral in K_k. This proves (31) on every polynomial test. Density and continuity give the identity in $\mathcal{D}^*$.

Define nonnegative constants recursively by

$$B_0 = 1, \qquad B_{r+1} = pT\sum_{k=0}^{\min(r,p-1)}\binom{p-1}{k}(r)_k B_{r-k}. \tag{32}$$

They depend on p, T, r, not on the mesh or H.

Lemma 8.1. *For a strict-past polynomial H and a cell of normalized length $h \le 1$,*

$$\left\| J_T(Hx_j^r) \right\| \le B_r \left\| J_T(H) \right\|, \tag{33}$$

$$\left\| J_T(Hx_j^q) \right\| \le B_q h \left\| J_T(H) \right\| \quad (q \ge 1), \tag{34}$$

$$\left\| J_T(Hx_j^q) \right\| \le B_q h^2 \left\| J_T(H) \right\| \quad (q > p). \tag{35}$$

For $1 \le q \le p$,

$$\left\| J_T(Hx_j^q) - I_q(W_j H) \right\| \le B_q h^2 \left\| J_T(H) \right\|. \tag{36}$$

Proof. The channel bound for a step is $\|K_k(W_j V)\| \le pT\binom{p-1}{k} h \|J_T(V)\|$. Apply it in (31) and induct on $r+1$, using (32). This first proves (34); since $h \le 1$, it also proves (33) and closes the induction. For $q > p$, every surviving index $k \le p-1$ has $q-1-k \ge 1$. Use (34) on that residual power in each term of (31). The additional step length supplies h^2, proving (35). For $q \le p$, the last term $k = q-1$ is exactly $(q-1)!K_{q-1}(W_j H) = I_q(W_j H)$. Every other term has a positive residual power, so the same argument bounds their sum by $B_q h^2 \|J_T(H)\|$. □

By (4), $\mu_q(Th) = 0$ for $1 \le q < p$ and $\mu_p(Th) = cp!Th$. For $q > p$, $|\mu_q(Th)| \le M_q h^2$ with a mesh-independent M_q: if $q = pk$, take $M_q = q!T^k/k!$ and note $k \ge 2$; otherwise take zero. Thus summing over $N = 1/h$ cells, under any uniform bound $\|J_T(H_j)\| \le M$, gives the following three limits:

$$J_T\left(\sum_j H_j (x_j^q - \mu_q(Th)) \right) \longrightarrow I_q(A), \qquad 1 \le q < p, \tag{37}$$

$$J_T\left(\sum_j H_j (x_j^p - \mu_p(Th)) \right) \longrightarrow 0, \tag{38}$$

$$J_T\left(\sum_j H_j (x_j^q - \mu_q(Th)) \right) \longrightarrow 0, \qquad q > p, \tag{39}$$

where the first line also assumes $\sum_j W_j H_j \to A$ in $\mathcal{F}$. The errors are $O(h)M$ in dual norm. For the critical line, (23) gives $I_p(W_j H_j) = cp!ThJ_T(H_j)$, which cancels the centering exactly. This is why the final centered reconstruction has $p-1$ channels.

9 Causal coefficients and convergence under refinement

It remains to prove the field convergence needed in (37). Introduce the causal comparison coefficient

$$b_{j,q}(P) = \frac{1}{q!} \Pi_j (\partial_{x_j}^q P), \tag{40}$$

and its step field $\widetilde{A}_{n,q} = \sum_j W_j (b_{j,q}(P_n))$. The derivative here is taken before the strict-past projection. It does not in general commute through that projection.

Lemma 9.1. *For every grid polynomial,*

$$b_{j,q}(P) = \frac{1}{q!} R_j e^{cTh\partial_{x_j}^p} \partial_{x_j}^q \Pi_{j+1} P. \tag{41}$$

Consequently, for a fixed cylinder under refinement,

$$\left\|\widetilde{A}_{n,q} - A_{n,q}\right\|_{\mathcal{F}} \le \frac{\|J_T\|\, e^T}{q!}(e^{Th} - 1)\,\|P\|_{\mathcal{D}} \longrightarrow 0. \tag{42}$$

Proof. The current coordinate is retained by Π_{j+1}, so its derivative commutes with the future integrations. Integrating that one current coordinate after future conditioning is exactly heat evolution followed by clamping, by (5). This proves (41). The norm of $e^{cTh\partial_{x_j}^p} - \mathrm{id}$ is at most $e^{Th} - 1$. Use contractivity of derivatives and clamping, the e^T projection bound, and the sum of normalized cell lengths equal to one. □

The heat correction proves equality of the eventual native and causal limits, but does not itself prove that either sequence has a limit. For that we need control of a moving conditioning cut.

Lemma 9.2. *For each degree bound d, there is $L_d \ge 0$, depending only on p, T, d, such that on every grid and every degree-at-most-d polynomial P,*

$$\|J_T(\Pi_b P - \Pi_a P)\| \le L_d(b-a)h\,\|P\|_{\mathcal{D}}, \qquad 0 \le a \le b \le N. \tag{43}$$

Proof. For each $q \ge 1$ there is $V_q \ge 0$ with $|\mu_q(Th)| \le V_q h$: use zero if $p \nmid q$, and $V_q = q!T^k/k!$ if $q = pk$, since $h^k \le h$. The cell identity (27), the positive power estimate, and (26) imply

$$\|J_T(\Pi_{j+1}P - \Pi_j P)\| \le h\,\|P\|_{\mathcal{D}}\,\|J_T\|\,e^T \sum_{q=1}^{d} \frac{B_q + V_q}{q!}.$$

Take the finite nonnegative sum as L_d, and sum the cell differences from a to $b-1$ by the triangle inequality. The argument proves dual-norm continuity of conditional increments; it does not assert continuity of moving-time positions in the strong factorial source norm. □

Proposition 9.3. *For each fixed cylinder P and every $q \ge 0$, the native fields $A_{n,q}(P)$ converge in $\mathcal{F}$ to a field C_qP. The causal fields converge to the same limit, and the bound* (1) *holds on the core.*

Proof. Compare adjacent grids. Each coarse increment splits into two fine increments. Differentiating the pulled-back polynomial in either fine coordinate is the pullback of differentiating in the coarse coordinate. Its degree stays at most $d = \deg P$, and its factorial norm after any number of such derivatives is at most $\|P\|_{\mathcal{D}}$.

At the first half of a coarse cell, the strict-past projections agree by signed convolution consistency. At the second half, the fine conditioning cut has advanced by just one fine cell. Apply (43) to the derivative polynomial. Dividing by $q!$ and integrating over the unit normalized horizon gives

$$\left\|\widetilde{A}_{n+1,q} - \widetilde{A}_{n,q}\right\|_{\mathcal{F}} \le \frac{L_d}{q!}2^{-(n+1)}\,\|P\|_{\mathcal{D}}. \tag{44}$$

This bound is summable in n. Thus the causal fields are Cauchy in the complete space $\mathcal{F}$ and converge. Equation (42) gives the same limit for the native fields. The norm bound follows by passing to the limit in (30). Each native field is an adapted polynomial step field; closedness gives $C_qP \in \mathcal{A}_T$. □

For $q > d$ all native fields are zero, hence $C_qP = 0$. If two base-grid presentations define the same cylinder, their refinements agree on a common tail. Uniqueness of norm limits therefore makes the limit independent of the presentation. Finite-grid linearity and continuity of addition and scalar multiplication make $P \mapsto C_qP$ linear. These observations are necessary before extending the maps to completed observables.

10 The multichannel formula and observable completion

Proposition 10.1. *Every polynomial cylinder satisfies*

$$J_T(P - \mathcal{E}_T P) = \sum_{q=1}^{p-1} I_q(C_q P).$$

Proof. Refine the fixed polynomial to all sufficiently fine grids. Choose $d' \geq \max(\deg P, p-1)$ once and pad the exact telescope (28) to degree d' with zero coefficients. Refinement preserves the degree bound, so this is the same finite number of terms on every grid. The left side, after applying J_T, is the same dual element on each grid.

For each q, (26) gives a uniform bound on the paired native coefficients. Proposition 9.3 provides the L^1 convergence for the low powers. Apply (37) to $q < p$, the exact critical cancellation (38) to $q = p$, and (39) to $q > p$. Pass through the one fixed finite sum. The surviving terms are exactly those in the displayed identity. □

Proof of Theorem 1.1. The degree-independent bound (1) on the dense core extends each linear coefficient map uniquely to a bounded linear map $C_q : \mathcal{D} \to \mathcal{F}$ with the same bound. For a completed observable choose polynomials $P_k \to u$ in $\mathcal{D}$. Then $C_q P_k \to C_q u$. Closedness of $\mathcal{A}_T$ preserves adapted membership. The centered map $u \mapsto J_T(u - \mathcal{E}_T(u)1)$ is bounded linear, as is each $I_q C_q$. Thus Proposition 10.1 passes to the limit and proves (3) for every $u \in \mathcal{D}$.

For the simultaneous actual signed approximation, assemble the $p-1$ native fields and the native surviving power sum into one finite product Banach space. For each fixed P_k, the preceding arguments show convergence of that package as the mesh tends to zero. Its limiting package is the collection $(C_1 P_k, \dots, C_{p-1} P_k, J_T(P_k - \mathcal{E}_T P_k))$, which tends to the analogous package for u by continuity. Choose a mesh index n_k at least k and at least the base-grid index of P_k, large enough that the package error is less than $1/(k+1)$. The triangle inequality gives simultaneous convergence to the package for u. Every approximating sum comes from the actual finite signed polynomial laws by (22). □

The diagonal is a proved selection; it is not every arbitrary joint degree/mesh limit. The convergence constants for one fixed polynomial may depend on its degree, as L_d shows. The extension to $\mathcal{D}$ instead uses the degree-independent coefficient bound. Those two kinds of bounds have distinct roles in the proof.

11 Uniqueness and the biharmonic case

The lower derivative orders cannot all be replaced by a coefficient depending only on the pth derivative. The terminal cylinder $P = Y_1$ has $D_s^p P = 0$ for every s, and $\mathcal{E}_T P = 0$. Nevertheless

$$J_T(P)(Y_1^{p-1}) = \mathcal{E}_T(Y_1^p) = cp!T \neq 0.$$

Thus any proposed reconstruction which vanishes whenever the pth derivative vanishes fails on this actual signed polynomial pairing. This elementary obstruction explains the role of lower channels without invoking a positive pseudo-L^2 interpretation.

If another bounded linear coefficient map agrees with C_q on all canonical polynomial limits, it equals C_q by density. This proves uniqueness of the coefficient extension. It does not assert that

an arbitrary channel representation has unique integrands. Let

$$S : (\mathcal{F})^{p-1} \to \mathcal{D}^*, \qquad S(A_1, \dots, A_{p-1}) = \sum_{q=1}^{p-1} I_q A_q.$$

Then $S(A) = S(B)$ if and only if $A - B \in \ker S$. No theorem here sets that kernel equal to zero.

For $p = 4$ the three surviving channels are

$$(I_1 A)(v) = -4T \int_0^1 A(s)(D_s^3 v)\, ds,$$
$$(I_2 A)(v) = -12T \int_0^1 A(s)(D_s^2 v)\, ds,$$
$$(I_3 A)(v) = -24T \int_0^1 A(s)(D_s v)\, ds.$$

The native coefficient fields retain factors 1, 1/2!, and 1/3! respectively. The fourth channel is the drift $I_4 = -24TQ_0$ and cancels against the fourth moment centering. This example keeps the signs, factorials, and complementary test derivative orders visible.

For $p = 2$ the same construction has one surviving channel $I_1 = 2TQ_1$, with the kernel normalization of generator ∂_x^2. It is compatible with that Brownian variance convention, but the theorem's proof has not relied on positivity or on importing a probability-space martingale representation.

12 Why the reconstruction stays in the test dual

The topology and channel orders in Theorem 1.1 are essential parts of its meaning. We give actual signed-law tests of stronger interpretations. Except for the final terminal pairing, these statements concern the native finite signed increment measures rather than the factorial analytic completion. No positive sampling law is inserted.

12.1 A finite-variation path-law obstruction

Assume $p = 2m > 2$. The moment formula (4) gives $\int x^2 k_\tau(x)dx = 0$, while $\int k_\tau = 1$. If $k_\tau \geq 0$ almost everywhere, the nonnegative integrable function $x^2 k_\tau(x)$ would vanish almost everywhere. Since Lebesgue measure has no atom at zero, this would force $k_\tau = 0$ almost everywhere, contradicting its unit mass. Thus the kernel is not nonnegative. Writing $k_\tau = k_\tau^+ - k_\tau^-$ shows

$$\int |k_\tau| = 1 + 2 \int k_\tau^- > 1.$$

Every integral is finite because the kernel is Schwartz.

The inverse Fourier integral gives the exact scaling

$$k_\tau(x) = \tau^{-1/p} k_1(x\tau^{-1/p}), \qquad A_p := \int_{\mathbb{R}} |k_\tau(x)|dx = \int_{\mathbb{R}} |k_1(x)|dx > 1. \tag{45}$$

Indeed substitute $\xi = \tau^{-1/p}\eta$ in its absolutely convergent inverse transform and then change variables in the absolute integral. Thus the absolute mass does not become small with the

time step. For N increments on a fixed horizon T, the actual signed product law $\mu_N(dx) = \prod_{j=1}^N k_{T/N}(x_j)dx_j$ satisfies, by Tonelli,

$$|\mu_N|(\mathbb{R}^N) = \prod_{j=1}^N \int |k_{T/N}| = A_p^N. \tag{46}$$

Suppose a countably additive signed measure ν of finite total variation on any measurable space had measurable N-increment observations with these pushforward laws for every N. For every measurable map F, $|F_\#\nu|$(whole space) $\le |\nu|$(whole space). To prove this inequality, pull back any finite measurable partition used in the variation supremum; the sum of the absolute signed masses is bounded by the total variation of ν. Applying it to these observations would give $A_p^N \le |\nu|$(whole space) for all N, impossible. In particular there is no such finite signed path law on $[0, T]$ realizing all the indicated uniform-grid increment distributions. No path continuity or consistency of different observation maps was needed for this contradiction.

The formal path-obstruction declaration uses arbitrarily many successive increments at a fixed positive step, hence increasing horizons. The fixed-horizon refinement conclusion just proved is an explicit classical deduction combining the same variation contraction with exact kernel scaling. It is not misattributed to a different formal path declaration. For $p = 2$, $A_2 = 1$ and this obstruction does not apply. It concerns finite total variation; it does not preclude generalized integration or the bounded test-dual construction proved above. The classical obstruction is also discussed in the primary introduction of [3].

12.2 Unbounded means on continuous cylinder unit balls

The variation obstruction also has a direct functional form which does not presume a measure representation theorem. Fix $p = 2m > 2$ and $T > 0$. Let $\mathcal{C}$ be the algebra of bounded continuous dyadic cylinder functions on continuous zero-start paths, with their path supremum norm. This is a different test class from the unbounded polynomial cylinders used to construct $\mathcal{D}$. Its finite signed expectation is well defined by the same actual increment laws and their refinement consistency. Then for every real R there is $F \in \mathcal{C}$ with

$$\|F\|_\infty \le 1, \qquad \mathcal{E}_T F > R. \tag{47}$$

Consequently no bounded linear functional on this cylinder algebra, its sup-norm closure, or the whole bounded-continuous path space can agree with all these expectations.

Here is a proof with actual continuous tests. On a grid of $N = 2^n$ increments of duration $h = T/N$, write $\rho_N(x) = \prod_{j=1}^N k_h(x_j)$ on $\mathbb{R}^N$. This density is continuous and integrable, and

$$\int_{\mathbb{R}^N} |\rho_N(x)|\, dx = A_p^N, \qquad A_p > 1$$

by the scaling and product calculation above. For $\varepsilon > 0$ the soft sign

$$g_\varepsilon(x) = \frac{\rho_N(x)}{|\rho_N(x)| + \varepsilon}$$

is continuous and bounded in absolute value by one, and

$$\int \rho_N g_\varepsilon = \int \frac{\rho_N^2}{|\rho_N| + \varepsilon} \longrightarrow \int |\rho_N| = A_p^N.$$

The integrand tends pointwise to $|\rho_N|$, including at its zeros, and is dominated by that integrable function, which proves the limit. Choose a dyadic N with $A_p^N > R$ and then a sufficiently small ε with the displayed integral greater than R. Compose g_ε with the continuous map taking a path to its successive grid increments. The resulting F is an actual bounded continuous cylinder with norm at most one. Its signed expectation is the displayed integral, proving (47). If a bounded agreeing functional had norm C, choose $R > C$ to contradict its unit-ball bound.

This is a sup-norm obstruction on bounded continuous tests, distinct both from the finite-variation path-measure obstruction and from the later compact-open mean nonclosability example on unbounded polynomials. It does not contradict bounded reconstruction on the specified factorial domain. At Gaussian order $p = 2$, $A_p = 1$ and this growth argument does not apply.
The soft-sign and native cylinder proofs are in `ContinuousDensityDuality`.
The exact actual-path unit-ball theorem is `HigherHeatUnboundedExpectation.unbounded_unit_cylinders`; its `no_uniform_bound` is the stated norm obstruction. Both source modules were read completely. This is the classical total-variation mechanism expressed directly on continuous tests; no new general functional-analysis principle is claimed.

12.3 Signed quartic cancellation versus absolute error

Take the biharmonic order $p = 4$, so $c = -1$, and put $h = T/N$. For the actual law μ_N above define $Q_N(x) = \sum_{j=1}^N x_j^4$. Its pointwise value is nonnegative, but its signed mean is

$$\int Q_N \, d\mu_N = N(-24h) = -24T. \tag{48}$$

The one-increment moments are $\int x^4 k_h = -24h$ and $\int x^8 k_h = 8!h^2/2 = 20160h^2$. Product integration, justified by absolute polynomial integrability, therefore gives

$$\begin{aligned} \int Q_N^2 \, d\mu_N &= 20160Nh^2 + 576N(N-1)h^2 \\ &= 576T^2 + 19584T^2/N. \end{aligned}$$

Expanding the centered square and using (48) yields the exact cancellation

$$\int (Q_N + 24T)^2 \, d\mu_N = 19584T^2/N \longrightarrow 0. \tag{49}$$

This is a signed integral of a square, not a norm-convergence statement. In fact $Q_N + 24T \geq 24T$ everywhere. Consequently the actual absolute error has the opposite behavior:

$$\int |Q_N + 24T|^2 \, d|\mu_N| \geq 576T^2 A_4^N \longrightarrow \infty. \tag{50}$$

Each integral is finite, since the product variation has every polynomial moment. The absolute error even has the lower bound $576T^2$ under the normalized variation probability $|\mu_N|/A_4^N$. Thus neither ordinary probability convergence to $-24T$ under those probability laws nor an L^2 isometry can be read from (49). The signed identity and the divergent variation error concern precisely the same native sums.

12.4 An L^2-small deterministic control with a fixed law

The proper control scale can be tested before any completed observable calculus. If a is constant on finitely many disjoint time intervals of lengths $h_j > 0$, define its biharmonic signed integral law by the pushforward of the product increment law under $\sum_j a_j x_j$. Multiplying the native Fourier transforms proves

$$\int e^{i\xi y}\, d\mathcal{L}_a(y) = \exp\left(-\xi^4 \sum_j h_j a_j^4\right).$$

The sum is the fourth-power control clock, not its second-power norm. On $[0,1]$ take the actual step functions

$$a_n(s) = (n+1)1_{(0,(n+1)^{-4}]}(s), \qquad n \geq 0.$$

They belong to both L^2 and L^4, and exactly

$$\|a_n\|_2 = (n+1)^{-1} \longrightarrow 0, \qquad \int_0^1 a_n^4 = 1.$$

Thus $\widehat{\mathcal{L}_{a_n}}(1) = e^{-1}$ for every n, whereas the zero control gives the Dirac law with Fourier value one. Any scalar functional on L^2 agreeing with these native Fourier values and equal to one at zero is discontinuous at zero, by this explicit convergent sequence. A weakly continuous law extension agreeing with the native laws is likewise impossible, since the real and imaginary parts of e^{iy} are bounded continuous tests. This concerns a plain L^2 input norm for the fourth-order integral; it does not contradict the factorial observable-domain theorem or an extension on a stronger fourth-power control norm.

The formal source uses Fourier phases $e^{2\pi i\xi y}$, so its unit-frequency witness is $e^{-(2\pi)^4}$ rather than e^{-1}. We use the angular-frequency convention fixed at the start of this paper. These are the same physical signed laws at correspondingly rescaled frequencies, with the same strict failure of continuity.

12.5 The last complementary derivative channel is essential

Return to the analytic completion and assume $p > 2$. Let X be the terminal coordinate polynomial at horizon T. On the polynomial core its signed mean and pairing satisfy

$$\mathcal{E}_T(X^{p-1}) = 0, \qquad J_T(X^{p-1})(X) = p!cT \neq 0.$$

Both follow directly from (4). For $1 \leq q \leq p-2$ one has $p - q \geq 2$. Since X is linear, $D_s^{p-q}X = 0$ for every s, and the channel definition (2) gives

$$(I_q A)(X) = 0 \qquad \text{for every admissible field } A.$$

Every sum of those channels annihilates X, whereas the centered pairing of X^{p-1} does not. Hence those channels alone cannot reconstruct that observable. In particular the first channel alone fails. In the full $p-1$ formula, only its last channel can contribute to this particular terminal test. This proves necessity of the last complementary-derivative channel for the displayed reconstruction; it does not prove an arbitrary general minimality theorem or injectivity of channel synthesis. The collective obstruction is also formalized in `HigherHeatMultichannelSharpness.all_lower_channels_insufficient`. The source uses zero-based channel indexing $r = q - 1$.

12.6 Compact-open path convergence does not close the mean

The norm in Section 4 is substantive. Consider instead $\mathcal{P} = \{\omega \in C(\mathbb{R}, \mathbb{R}) : \omega(0) = 0\}$ and the space $C(\mathcal{P}, \mathbb{R})$ with uniform convergence on compact subsets of $\mathcal{P}$. Dyadic polynomial cylinders have a faithful realization in this space. Indeed each dyadic evaluation map is continuous; every finite list of prescribed values is realized by a piecewise linear path starting at zero. Equality of two realized cylinder polynomials therefore gives equality of the finite polynomials after passage to a common refinement. Thus the original cylinder mean defines an unambiguous linear functional on this realized polynomial algebra. The path coordinate is normalized time: $U(\omega) = \omega(1)$ has the original terminal signed moments at physical horizon T.

Set $a = cT \neq 0$ and, for $n \geq 0$, form the actual cylinder

$$F_n(\omega) = \frac{n!}{(pn)!a^n} U(\omega)^{pn}.$$

Equation (4) gives $\mathcal{E}_T(F_n) = 1$ exactly. Nevertheless $F_n \to 0$ uniformly on every compact subset of $\mathcal{P}$. To prove this, for compact $K \subset \mathcal{P}$, continuity of U gives $|U| \leq R$ on K for a finite $R \geq 0$. The integer inequality $(pn)! \geq (n!)^2$ holds for $p \geq 2$: $\binom{2n}{n} \geq 1$ gives $(2n)! \geq (n!)^2$ and factorials increase from $2n$ to pn. Consequently

$$\sup_{\omega \in K} |F_n(\omega)| \leq \frac{(R^p/|a|)^n}{n!} \longrightarrow 0.$$

For the limit, if $A = R^p/|a|$, the consecutive ratio of $A^n/n!$ is $A/(n+1)$, eventually at most $1/2$; the zero case is immediate. This proves the compact-open convergence.

Proposition 12.1. *The realized polynomial mean has no closed linear partial-operator extension from $C(\mathcal{P}, \mathbb{R})$ to $\mathbb{C}$ with the compact-open input topology. In particular its graph is not closable in that topology.*

Proof. Its graph contains $(F_n, 1)$, which converges to $(0, 1)$ by the proved uniform-on-compacts limit. Every closed graph containing the polynomial graph must therefore contain $(0, 1)$. The graph of a linear operator also contains $(0, 0)$ and has only one value at zero. This is impossible. In particular the closure of the original graph cannot be the graph of a linear partial operator. □

This obstruction holds already at $m = 1$, where the kernel is the positive Gaussian heat law of variance $2T$. It therefore does not prove absence of a probability representation. Compact-open control of unbounded polynomials does not control their integrals, even in that positive case. It also does not contradict the bounded mean on the stronger factorial analytic completion: this sequence is not asserted to converge in the norm of that completion. No claim about expectations of uniformly bounded path functions follows from this particular sequence.

12.7 One starting point does not separate Schwartz tests

A different testing distinction survives even at fixed positive time. The nonzero real Schwartz function $f(x) = xe^{-x^2}$ is odd, whereas every k_t in this paper is even. Their product is integrable: f is bounded and $k_t \in L^1$, or one may use Schwartz decay. Changing x to $-x$ then gives

$$\int_{\mathbb{R}} f(x)k_t(x)\,dx = 0 \qquad \text{for every } t > 0, \qquad f(1) = e^{-1} > 0.$$

Thus all zero-start heat marginals together do not separate the Schwartz class. This is an elementary parity obstruction, not a failure of Fourier injectivity or a proof that the completed analytic pairing J_T is noninjective.

In fact all starting points at one fixed positive time do separate Schwartz functions. If $f \in \mathcal{S}(\mathbb{R}, \mathbb{C})$ satisfies

$$\int_{\mathbb{R}} f(x+y)k_t(y)\,dy = 0 \qquad \text{for every } x \in \mathbb{R},$$

then evenness of k_t identifies this function with $k_t * f$. Both factors are Schwartz, and the convolution Fourier formula gives

$$e^{-t\xi^p}\widehat{f}(\xi) = 0 \qquad (\xi \in \mathbb{R})$$

in the angular-frequency convention of Section 1. The multiplier never vanishes, so $\widehat{f} = 0$ pointwise and Fourier inversion gives $f = 0$. This proves separation without a bounded inverse: dividing by $e^{-t\xi^p}$ need not preserve the Schwartz class for an arbitrary proposed output. These topology and testing obstructions use classical factorial and Fourier mechanisms; no new general extension principle is claimed.

12.8 A separate Schwartz observable identity for actual L^p controls

The L^2 discontinuity test of Section 12.4 does not exclude a control theory in the correct power norm. The frozen library also constructs the following deterministic-control identity. It uses Schwartz terminal tests, separately from the factorial multitime completion in Theorem 1.1; no inclusion of these domains is assumed. Fix $T \geq 0$, $p = 2m$, $c = (-1)^{m+1}$ and a real $u \in L^p([0,T])$. Set

$$A_u(t) = \int_0^t |u(s)|^p ds \quad (0 \leq t \leq T), \qquad \nu_{u,t} = \begin{cases} k_{A_u(t)}(x)\,dx, & A_u(t) > 0, \\ \delta_0, & A_u(t) = 0. \end{cases}$$

These are actual finite signed measures of mass one. The clock is continuous, nondecreasing and absolutely continuous, with derivative $|u(t)|^p$ almost everywhere. Its definition is unchanged when u is replaced by an equal almost-everywhere representative. For $f \in \mathcal{S}(\mathbb{R}, \mathbb{C})$ write $\mathsf{E}_u(t)f = \int f\,d\nu_{u,t}$, an actual signed integral.

Proposition 12.2. *For every such u, f the scalar curve $t \mapsto \mathsf{E}_u(t)f$ is absolutely continuous and satisfies*

$$\mathsf{E}_u(T)f - f(0) = \int_0^T |u(t)|^p c\,\mathsf{E}_u(t)f^{(p)}\,dt.$$

The drift is absolutely integrable, bounded in modulus by $M_f|u(t)|^p$, where in angular frequencies

$$M_f = \frac{1}{2\pi}\int_{\mathbb{R}} |\xi|^p |\widehat{f}(\xi)|\,d\xi < \infty.$$

The derivative identity holds almost everywhere, including times of zero accumulated clock.

Proof. Schwartz Fourier inversion and the actual kernel transform give, for $r \geq 0$ including $r = 0$,

$$F_f(r) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{f}(\xi)e^{-r\xi^p}d\xi, \qquad \mathsf{E}_u(t)f = F_f(A_u(t)).$$

For $r > 0$ the signed-kernel equality is justified by absolute Fubini, since f is bounded, the kernel is integrable and $\widehat{f} \in L^1$. At $r = 0$ Fourier inversion gives $F_f(0) = f(0)$, agreeing with δ_0.

Differentiating under the frequency integral gives a continuous right derivative at zero and an ordinary continuous derivative for $r > 0$:

$$F_f'(r) = -\frac{1}{2\pi}\int \xi^p \widehat{f}(\xi) e^{-r\xi^p} d\xi = cF_{f^{(p)}}(r), \qquad |F_f'(r)| \le M_f.$$

Domination uses $|\xi|^p|\widehat{f}(\xi)|$, valid for every $r \ge 0$; the sign follows from $c(i\xi)^p = -\xi^p$. Extend F_f to negative arguments by $F_f(r) = f(0) + rF_f'(0)$ for $r < 0$. This is a genuine C^1 function on $\mathbb{R}$, with derivative bounded by M_f, agreeing with the heat expectation for nonnegative arguments. It is an auxiliary linear extension, not backward heat evolution.

The extended function is globally Lipschitz. Composition with the absolutely continuous clock is absolutely continuous; the chain rule at almost every clock differentiability point gives

$$\frac{d}{dt}\mathsf{E}_u(t)f = |u(t)|^p c\, \mathsf{E}_u(t) f^{(p)}.$$

The C^1 extension makes this chain rule valid also where the clock equals zero. The derivative bound gives an integrable majorant $M_f|u(t)|^p$. Applying the absolutely continuous fundamental theorem of calculus proves the integrated identity. □

Source correspondence: `HigherHeatLpControls.power_integrable`, `clock_integral`, `marginal_mass`; `HigherHeatLpTimeChange.time_continuous`, `time_monotone`. `HigherHeatObservableEndpoint.extension_hasDerivAt`; `HigherHeatLpEndpointIto.expectation_eq_mean`, `observable_ito`. `HigherHeatLpAbsoluteContinuity.expectation_ac`; `HigherHeatLpIntegratedIto.drift_integrable`, `integrated_ito`.

The native control space is $L^p(\mathbb{R})$; a control on $[0,T]$ is extended by zero. The source Fourier convention is unit frequency; the substitution $\xi = 2\pi\eta$ gives exactly the displayed angular constant M_f. No extra factorial or variance factor enters its generator $c\partial^p$.

There is also a useful quantitative continuity statement, printed here as an elementary human deduction. For $u, v \in L^p([0,T])$,

$$\sup_{0\le t\le T} |\mathsf{E}_u(t)f - \mathsf{E}_v(t)f| \le M_f p(\|u\|_p + \|v\|_p)^{p-1}\|u - v\|_p.$$

Indeed the Lipschitz bound reduces the left side to $M_f \int_0^T ||u|^p - |v|^p|$. The real mean value theorem gives $|a^p - b^p| \le p(a+b)^{p-1}|a-b|$ for $a, b \ge 0$. Use $||u| - |v|| \le |u - v|$, Hölder with exponents $p/(p-1)$ and p, and the L^p triangle inequality to obtain the claimed estimate. This proves convergence of the signed Schwartz observable curves when the controls converge in L^p.

For the earlier biharmonic spikes a_n, one has $\|a_n\|_4^4 = 1$, while $\|a_n\|_2 \to 0$. They therefore do not converge to zero in the control norm used here, and their fixed fourth-power clock is consistent with this continuity theorem. This is a deterministic absolutely continuous clock and a scalar signed expectation identity. It supplies no probability path law, random signed-clock composition, arbitrary predictable control integral, or extension of the factorial multichannel coefficients to all Schwartz observables. The semigroup and chain-rule mechanisms are classical; no new stochastic integration principle is claimed.

13 Verified source and literature boundary

The mathematical source is the frozen Git commit

`bab688bb6140cd6316c80680095a914ea0d346db`.

The namespaces in the source map share the following prefix:

`OperatorDerivativeRewrite.Foundation`

The file `theorem-map.json` records exact files, declarations, line numbers, source hashes and baseline GitHub links.

Content	*Lean namespace and principal declaration*
Domain and core	`HigherHeatAnalyticCylinderDomain.core_injective`, `core_dense`; `MultivariateFactorialDomain.derivative_norm`
Signed product pairing	`HigherHeatCompletedPairing.pairing_core_complex`, `pairing_not_positive`
Exact finite telescope	`HigherHeatMultitimeJetTelescope.telescope`, `native_dual_identity`
Power bounds	`HigherHeatAllPowerBounds.positive_increment_bound`, `higher_increment_bound`; `HigherHeatAllChannelRemainder.cell_remainder_bound`
Centered channel limits	`HigherHeatCenteredPowerLimits.low_centered_converge`, `critical_centered_vanish`, `higher_centered_vanish`
Native/causal correction	`HigherHeatCausalJetIdentification.physical_shifted_jet_eq`, `causal_field_approximation`
Cut and refinement bounds	`HigherHeatConditionalCutBound.conditional_cut_bound`; `HigherHeatCausalFieldCauchy.causal_fields_step_bound`
Field limits	`HigherHeatMultitimePredictableFields.field_tendsto`, `field_bound`
Polynomial reconstruction	`HigherHeatMultitimeReconstruction.polynomial_reconstruction`
Observable extension	`HigherHeatMultitimeFieldMaps.completed_bound`, `completed_mem`; `HigherHeatCompletedClarkOcone.clark_ocone`
Uniqueness	`HigherHeatCompletedClarkOcone.coefficient_unique`, `synthesis_uniqueness`
Actual joint diagonal	`HigherHeatSignedDiagonalRealization.completed_signed_realization`

The formal map also records the faithful one-coordinate realization, exact derivative recovery, actual theta moments and interval lower bounds, and the resulting nonmembership theorem. Its elementary series argument is a human reconstruction of that faithful realization, not a larger analytic domain or a new arithmetic theorem.

The baseline receipt records a complete successful root build and a complete 54,943-declaration standard-axiom audit. Publication work changes no Lean source and reuses that verified baseline. The human proofs above reorganize the actual theorem mechanism; the notation B_r, L_d corresponds to recursive power bounds and finite degree-dependent constants in the formal sources. It does not claim a fresh Lean theorem merely because a constant has a shorter name in this exposition.

The primary comparison in [3], especially its discussion leading to Theorem 18, already retains lower power increments in a generalized Itô calculus for complex random-walk approximations. Its primary introduction also explicitly discusses failure of ordinary signed path-measure realization;

our finite-variation obstruction is classical context, with all actual kernel/scaling details proved here. Its expected chain-rule formula is distinct from the present completed real signed-cylinder test-dual identity. That distinction is a comparison of the stated constructions, not a proof of priority for the present one. A closer antecedent is Nishioka [2]. His Section 3 defines weak limits by pairing against every Fourier-measure tame test at every starting point; these limits are linear functionals, not path functions. Theorem 3.1 constructs integrals, products and iterated integrals. In dimension one, Remark 3.3(iii) retains powers $1, \dots, p-1$ for $p = 2q$; Theorem 6.9 gives a weak Itô formula. Theorem 5.3 and Remark 5.4 identify complementary differential orders up to $p-1$. Thus neither the lower channels, their test-functional interpretation nor that differential correspondence is claimed new. His weighted Fourier-measure test spaces and Lemma 3.4 continuity estimates differ from the present factorial coefficient completion. Section 4.1 proves explicit one-coordinate noninclusions; comparison of the multitime completions and reconstruction topologies remains open. No priority follows from these differences. Hochberg's full text was not retrieved, so its direct comparison remains pending. Cinque–Orsingher [4], Section 3.1, equations (3.4)–(3.7), explicitly use a cylinder algebra and a pseudo-measure rather than an ordinary signed measure. Their opening infinite absolute-integral setting differs from our finite one-time Schwartz-kernel variation; our variation obstruction instead grows under refinement. Definition 3.1(vii) defines a time change through density composition when the defining integral converges. These distinctions preclude treating our finite-variation obstruction as a refutation of that framework, or importing its terminology as a probability filtration. Only the stated definitions and scope were compared here; its composition theorems and proofs were not independently audited. It supplies no premise extending our completed reconstruction to signed random clocks. Comparison of domains and reconstruction topologies remains necessary before a publication decision.

References

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[6] G. N. Hile and A. Stanoyevitch, *Heat Polynomial Generalizations*, author-hosted primary manuscript, Sections 3–4, exponential generating functions. `https://stanoyevitch.net/papers/no14_w_Hile_AMRT_02.pdf`.